Ex:no-4 Setting up a Local Area Network (LAN)

Aim:-

To design, set up, and configure a LAN connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment Required:-

S.no	Equipment	Quantity	Specification
1).	Personal Computers(PCs)	4	Ethernet (NIC) enabled
2).	Network Switch	1	8/16/24 port, unmanaged / managed
3).	Ethernet Cables(Cat5e/Cat6)	4	Straight-through, RJ-45 connectors

Theory:-

A LAN is a network that connects computers & devices within a limited geographical area such as a home, office or laboratory, allowing them to share resources such as files, printers and internet connections.

A star topology is used in this experiment, where all end devices (PCs) are individually connected to a central network switch.

Network Diagram:-

The LAN was implemented using a star topology as shown below, with 4 PCs connected individually to a central switch.

IP Address Configuration Table:-

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa 0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

6. Implementation in Cisco Packet Tracer:-

Since physical hardware was limited, the LAN was also implemented & verified using Cisco Packet Tracer simulation software, following the steps below.

Step 1: Place the Devices

Step 2:- Cable the Devices

Step 3:- Assign IP Addresses to PCs

Step 4:- Configure the Switch Management IP (Optional)

Step 5:- Verify Connectivity Using Ping

Step 6:- Demonstrate Data Sharing (Server Simulation)

Step 7:- Observe Packet Flow in Simulation Mode.

Observation:-

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet (10.0.0.0/255.0.0.0). Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.